

# Zipty Tools and Technologies

Here is the technology and tools stack used to build **Zipty5**:

## Core platform

| Category        | Technologies            |
|-----------------|-------------------------|
| Language        | TypeScript, JavaScript  |
| Runtime         | Node.js 22.x            |
| Framework       | Next.js 16 (App Router) |
| Frontend        | React 19                |
| Package manager | pnpm 10.x               |
| Version control | Git (GitHub)            |

## Frontend / UI

| Category              | Technologies                                                                   |
|-----------------------|--------------------------------------------------------------------------------|
| Styling               | Tailwind CSS 4, PostCSS, Autoprefixer                                          |
| UI components         | Radix UI, shadcn/ui (New York style)                                           |
| Icons                 | Lucide React, React Icons                                                      |
| Animations            | Framer Motion, React Spring, Tailwind Animate                                  |
| Forms                 | React Hook Form, Zod, Hookform Resolvers                                       |
| State / data fetching | TanStack React Query                                                           |
| Rich text editor      | Tiptap                                                                         |
| Charts                | Recharts                                                                       |
| Carousels / sliders   | Swiper, Embla Carousel                                                         |
| Other UI              | Sonner (toasts), Vaul (drawer), cmdk, input-otp, react-day-picker, next-themes |

## Backend / API

| Category                    | Technologies                                    |
|-----------------------------|-------------------------------------------------|
| <b>API layer</b>            | Next.js API Routes ( <code>/api/v1/*</code> )   |
| <b>Validation</b>           | Zod                                             |
| <b>Auth</b>                 | Custom JWT (jsonwebtoken), bcryptjs             |
| <b>Database client</b>      | <code>pg</code> (PostgreSQL)                    |
| <b>BaaS / DB platform</b>   | Supabase ( <code>@supabase/supabase-js</code> ) |
| <b>Queue / jobs</b>         | BullMQ                                          |
| <b>Cache / Redis client</b> | ioredis, Redis                                  |
| <b>Email</b>                | Nodemailer (SMTP)                               |
| <b>HTML sanitization</b>    | sanitize-html                                   |

## Database & schema

| Category                  | Technologies                                                               |
|---------------------------|----------------------------------------------------------------------------|
| <b>Database</b>           | PostgreSQL (hosted on Supabase)                                            |
| <b>Schema management</b>  | Raw SQL ( <code>supabase/sql/</code> , <code>supabase/migrations/</code> ) |
| <b>Connection pooling</b> | Supabase PgBouncer / transaction pooler                                    |

## Payments & commerce integrations

| Category                    | Technologies                                           |
|-----------------------------|--------------------------------------------------------|
| <b>Payments (primary)</b>   | Razorpay                                               |
| <b>Payments (supported)</b> | Stripe SDK                                             |
| <b>Shipping / logistics</b> | Shippo API                                             |
| <b>SMS</b>                  | Configurable providers (custom SMS API, MSG91, Twilio) |

|                             |                     |
|-----------------------------|---------------------|
| <b>India SMS compliance</b> | DLT templates       |
| <b>Email</b>                | SMTP (any provider) |

## File handling & exports

| Category                  | Technologies                               |
|---------------------------|--------------------------------------------|
| <b>Uploads</b>            | Local VPS filesystem + API upload pipeline |
| <b>ZIP handling</b>       | adm-zip                                    |
| <b>PDF generation</b>     | jsPDF, jsPDF-AutoTable                     |
| <b>Spreadsheet export</b> | xlsx                                       |

## Infrastructure & hosting

| Category               | Technologies                                    |
|------------------------|-------------------------------------------------|
| <b>Hosting</b>         | VPS                                             |
| <b>Reverse proxy</b>   | Nginx                                           |
| <b>Process manager</b> | systemd                                         |
| <b>CDN</b>             | AWS CloudFront ( <code>cdn.ziply5.com</code> )  |
| <b>Asset storage</b>   | VPS local disk ( <code>/var/www/assets</code> ) |
| <b>TLS / DNS</b>       | Domain + HTTPS at Nginx / load balancer         |

## DevOps & operations

| Category                 | Technologies                                                                |
|--------------------------|-----------------------------------------------------------------------------|
| <b>Build</b>             | Next.js build ( <code>next build</code> )                                   |
| <b>Git hooks</b>         | Husky                                                                       |
| <b>Backup scripts</b>    | Shell scripts ( <code>backup-db.sh</code> , <code>backup-assets.sh</code> ) |
| <b>Monitoring / logs</b> | systemd logs, custom shell checks                                           |

|             |                                         |
|-------------|-----------------------------------------|
| <b>Cron</b> | Linux crontab (backups, storage checks) |
|-------------|-----------------------------------------|

## Testing & API tooling

| Category                         | Technologies                                                                       |
|----------------------------------|------------------------------------------------------------------------------------|
| <b>Testing</b>                   | Node.js built-in test runner + tsx                                                 |
| <b>API testing</b>               | Postman (collections/globals in repo)                                              |
| <b>Smoke / migration scripts</b> | Custom Node scripts ( <code>seed-auth</code> , <code>smoke-supabase</code> , etc.) |

## Analytics & utilities

| Category               | Technologies                                   |
|------------------------|------------------------------------------------|
| <b>Analytics</b>       | Vercel Analytics                               |
| <b>Date handling</b>   | date-fns                                       |
| <b>Class utilities</b> | clsx, class-variance-authority, tailwind-merge |

## Application modules (functional areas)

These are not separate frameworks, but major parts of the app built on the stack above:

- Customer storefront (catalog, cart, checkout)
- Admin panel (dashboard, catalog, inventory, OMS)
- Order management system (OMS)
- Returns & refunds
- Inventory & warehouses (FIFO-ready)
- Promotions, coupons, offers
- CMS / content management
- Support tickets & notifications
- Reports & analytics snapshots
- Abandoned cart recovery
- OTP / user authentication
- Serviceability & pincode checks